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Vitamin B_6 nutrition, metabolism, and the relationship of diseases: current concepts and future research

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ABSTRACT

Vitamin B_6, a vital micronutrient acquired through dietary intake, plays a crucial role in numerous enzymatic reactions within the human body. Despite its significance, deficiencies in vitamin B_6 remain prevalent and are linked to a spectrum of chronic and acute diseases. This review explored the intricate relationships between vitamin B_6 metabolism and various diseases, focusing on cancer, diabetes, cardiovascular conditions, neurodegenerative disorders, and COVID-19-related complications. We highlighted the mechanistic roles of pyridoxal 5′-phosphate, the active form of vitamin B_6, in processes such as inflammation modulation, homocysteine regulation, and oxidative stress mitigation. By synthesizing recent advances in both clinical and preclinical studies, this paper underscores the therapeutic potential of vitamin B_6 while advocating for personalized nutritional interventions tailored to individual health profiles. Our findings aim to inform future research, foster targeted disease prevention strategies, and optimize the safe use of vitamin B_6 as part of a balanced nutritional approach.

1. Introduction

In recent years, global challenges posed by geopolitical conflicts, public health crises and environmental factors have contributed to the growing concern for food safety and nutritional balance[1]. Among the many micronutrients, vitamin B_6 (also known as pyridoxine (PN)), an important water-soluble vitamin, plays a key role in maintaining human health and preventing disease[2]. Vitamin B_6, which is ingested through diet or supplements, is a coenzyme in more than 160 enzymatic reactions involved in a variety of physiological processes, including amino acid metabolism, glucose synthesis, neurotransmitter production, and immune function[3-4]. Although the importance of vitamin B_6 in maintaining good health has long been recognized, vitamin B_6 deficiency remains a pressing issue in global public health, especially in the context of unbalanced diets and certain disease states where vitamin B_6 metabolism is compromised, leading to deficiency.

With the continuous development of science and technology, more and more studies have begun to reveal the role of vitamin B_6 in a variety of chronic diseases such as cardiovascular disease (CVD), diabetes, cancer and neurodegenerative diseases (NDDs)[5]. The potential of vitamin B_6 in the prevention, treatment and mitigation of these diseases has gradually attracted widespread attention in both academic and clinical medicine. Vitamin B_6 not only plays a role in regulating homocysteine metabolism, reducing oxidative stress, inhibiting inflammation, and promoting DNA repair, but also has an important impact on the maintenance and repair of the immune system and nervous system functions[6-7]. These functions of vitamin B_6 allow

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it to play multiple roles in disease prevention and intervention, and are particularly critical in the treatment of chronic diseases.

Despite the well-documented biological functions of vitamin B_6, vitamin B_6 deficiency remains a global health problem, especially among the elderly, immune-compromised and chronically ill[8]. Supplementation of vitamin B_6 based on individual health status, disease background, and nutritional needs has become a key issue in nutrition and public health.

According to the *Mayo Clinic guidelines* (2023), the recommended daily allowance (RDA) for vitamin B_6 is 1.3 mg/day for adults aged 19–50 years, regardless of gender. For adults aged 51 years and older, the RDA is 1.7 mg/day for males and 1.5 mg/day for females. While the dietary reference intakes primarily address the general population's nutritional needs, emerging research underscores the importance of reassessing supplementation strategies and understanding the mechanisms of vitamin B_6 in specific disease contexts. This review aims to investigate the role of vitamin B_6 in various diseases and synthesize current literature to inform future research directions and public health policies (Fig. 1).

2. Current status of knowledge

2.1 *Structure of vitamin B_6*

Vitamin B_6 serves as a coenzyme for amino acid decarboxylases and transaminases in protein metabolism and functions as a cofactor in over 160 enzymatic reactions[7]. It is involved in the metabolic processes of carbohydrates, lipids, amino acids, and nucleic acids, and also plays a role in regulating cell signaling[15]. Vitamin B_6 consists of 8 interconvertible isomers, including PN, pyridoxine 5′-phosphate, pyridoxal (PL), pyridoxal 5′-phosphate (PLP), pyridoxamine (PM), pyridoxamine 5′-phosphate (PMP), 4-pyridoxic acid (4-PA), and pyritinol, as shown in various studies[16-17]. In animal products[18], vitamin B_6 primarily exists as PL and PM, while plant-based sources contain higher amounts of glycosylated PN, known as pyridoxine-5′-*β*-*D*-glucoside, which has lower bioavailability than PN[19]. The structures of vitamin B_6 isomers are summarized in Fig. 2.

2.2 *Metabolism and physicochemical properties of vitamin B_6*

As a water-soluble vitamin, vitamin B_6 is most stable in acidic solutions, but it is susceptible to degradation when exposed to air, alkaline conditions, or UV radiation[20-21]. These physicochemical properties significantly influence its metabolism in the human body[22]. The overall metabolic process of vitamin B_6 in humans is presented in Fig. 3.

2.2.1 *Digestion and absorption of vitamin B_6*

Vitamin B_6 can be obtained from conventional foods, dietary supplements, and fortified foods, and is absorbed in the small intestine through passive diffusion[23]. In the lumen of the small intestine,

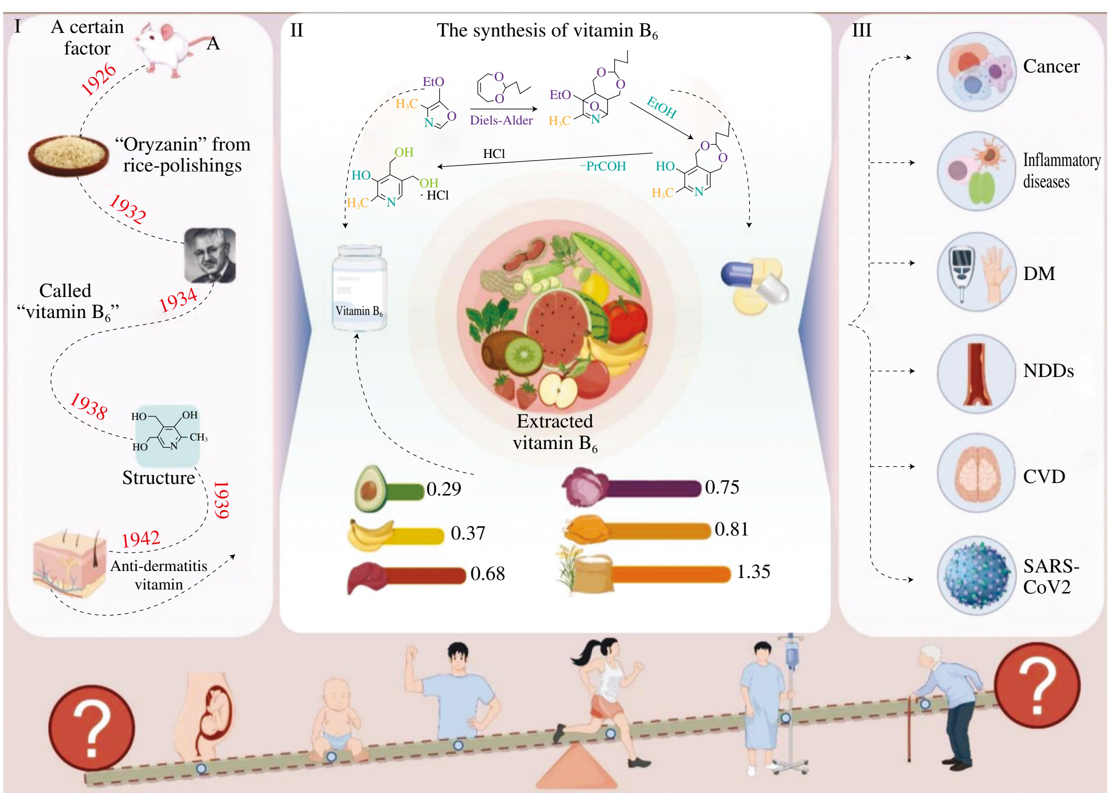

Fig. 1 Overview of vitamin B_6, from its discovery to its role in disease. I: The discovery of vitamin B_6[9]. Key milestones include identifying "Oryzanin" from rice polishings in 1926[10], naming it vitamin B_6 in 1934[11], elucidating its structure in 1938[12], and recognizing it as an anti-dermatitis vitamin in 1942[13]. II: The synthesis and sources of vitamin B_6[14]. This section highlights the chemical synthesis pathways and the best dietary sources of vitamin B_6 per 100 g, including foods like rice bran (1.35 mg), chicken (0.81 mg), liver (0.68 mg), and others. III: Diseases associated with vitamin B_6[2]. Vitamin B_6 is implicated in several conditions, such as cancer, inflammatory diseases, diabetes mellitus (DM), NDDs, CVD, and its role in combating severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).

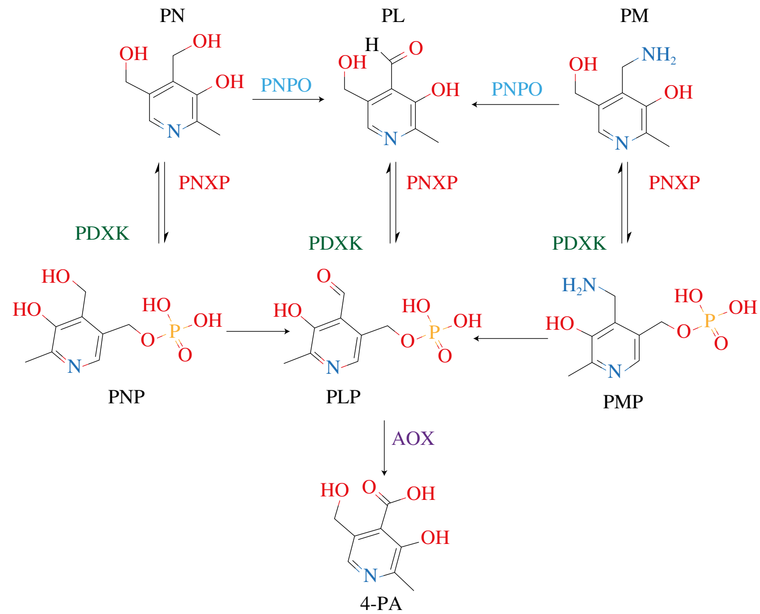

Fig. 2 The pathway of vitamin B_6 catabolism (both non-phosphorylated and phosphorylated forms), along with the enzymes and cofactors involved. PDXK, pyridoxal kinase; PNPO, pyridoxine 5′-phosphate oxidase; PNXP, pyridoxal 5′-phosphate, AOX, aldehyde oxidase; PNP, pyridoxine 5′-phosphate.

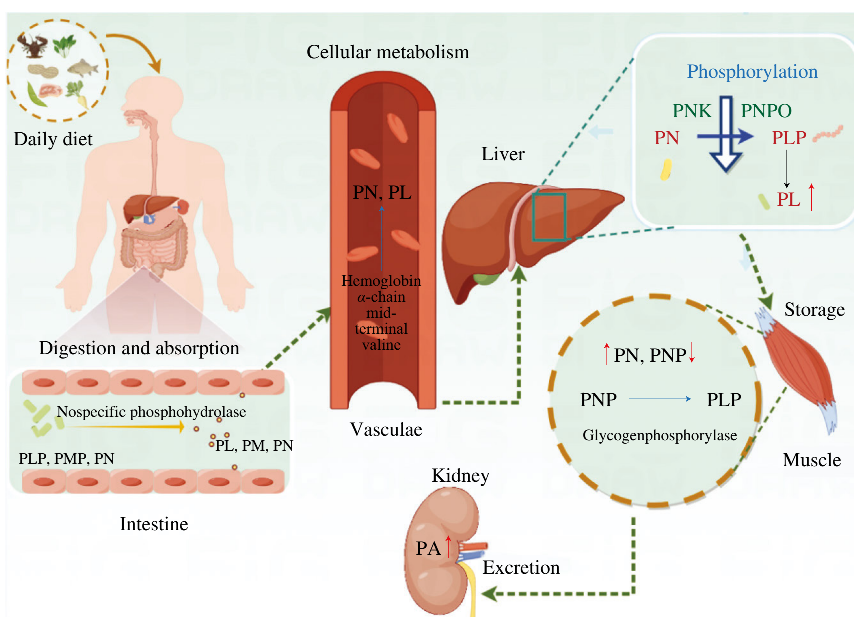

Fig. 3 Vitamin B_6 metabolism and its pathways. PNK, pyridoxal kinase.

PLP is hydrolyzed to PL, and PMP is converted to PM by a non-specific phosphatase. Subsequently, PL, PM, and PN are absorbed by intestinal cells. Studies have shown that a physiological dose of PM (ranging from 14 to 46 nmol per mouse) is rapidly converted to PL in the intestinal tissues, where it is released into the portal blood as PL[24-25]. Other research[26] indicates that dietary PN is converted to PL in circulation, which is then taken up and phosphorylated by other organs. Furthermore, most of the blood [3H] PL is found in the plasma, a localization that may facilitate its utilization by organs.

2.2.2 *Transport and cellular metabolism of vitamin B_6*

PN is transported to the small intestine mucosa and the bloodstream, and can also be synthesized in the intestinal mucosa, accounting for about 30.6% of the dose[27]. Approximately 10.4%–15.7% of the administered dose of PN in the bloodstream diffuses into the muscles, where it undergoes phosphorylation[28]. Plasma PL levels can increase 12-fold after PN administration in humans. Although PLP accounts for 60% of plasma vitamin B_6, it is protein-bound and not readily available to other cells[29]. Plasma PL, however, exhibits weak binding to albumin, is transportable, can be absorbed and eliminated by tissues, and is oxidized to PA. Both PN and PL enter erythrocytes by diffusion, where they are phosphorylated by kinases[30]. Human erythrocytes can oxidize PNP to PLP, a function absent in other animals, such as rats. PN enters erythrocytes within 3–5 min at concentrations above the erythrocyte PL kinase saturation point, resulting in an intracellular concentration equal to that of the medium. The amount of PL entering erythrocytes increases when concentrations surpass the erythrocyte phosphokinase concentration, making the intracellular concentration higher than in the medium[31]. This accumulation is attributed to the binding of PL to the terminal valine in the hemoglobin α-chain, causing PL to accumulate in erythrocytes. Consequently, PL levels in erythrocytes can be four to five times higher than in plasma, suggesting that erythrocytes may also function as a mode of transport.

The liver is another active site for vitamin B_6 metabolism. PN is incorporated into hepatocytes and transformed into PL through the actions of PL kinase and PNP oxidase. It is then converted into PLP in tissues with phosphokinase activity, which is transported to the circulatory system[32]. In brain sections and isolated choroid plexus, neither PN nor its phosphate compounds are found. The choroid plexus does not serve as a conduit for the transport of PN from blood to cerebrospinal fluid. In vitamin B_6-deficient rats[33-34], PL kinase activity decreased, with a 50% reduction in the liver after five weeks, compared to a 14% reduction in the brain. This highlights the importance of vitamin B_6 for the nervous system.

2.2.3 Storage of vitamin B_6

Vitamin B_6 is phosphorylated in the bloodstream upon diffusion into muscles. As the PN dose increases, the percentage of PN in muscle rises, while the percentage of PNP decreases[32]. In rats[35], no oxidation of PNP to PLP occurs in muscle tissue. About 60% of vitamin B_6 is found in muscle, 75%–95% of which is associated with glycogen phosphorylase, an enzyme that constitutes 5% of muscle soluble protein and may serve as a storage site for vitamin B_6. Muscle proteins are converted to meet the body's minimum requirements for vitamin B_6[33-34].

2.2.4 Excretion and absorption of vitamin B_6

PA, the primary metabolite of vitamin B_6, accounts for 20%–40% of total vitamin B_6 intake. PA levels in urine serve as an indicator of vitamin B_6 intake but do not reflect the amount of vitamin stored in the body. Additionally, small amounts of PN and PL are also found in urine[36]. PN accumulates in renal tubules and is excreted by the kidneys when concentrations are high. At a PN dose of 10 mg, PA excretion in urine decreases, while PN excretion increases. PL is not easily eliminated by the kidneys and accumulates in its phosphorylated form after incorporation. However, after oral administration of large doses (100 mg) of PL, PM, and PN in humans, most of the prodrugs are excreted in the urine within 36 h[37].

In rats, about 75%–80% of vitamin B_6 is stored in muscle tissue[38]. Most forms of vitamin B_6 are absorbed by passive diffusion in the jejunum and ileum, where they are phosphorylated to form PLP and PMP. The absorbed vitamin B_6 metabolites are bound to proteins in the intestinal mucosa and blood. Transport occurs via a non-saturated passive diffusion mechanism. Vitamin B_6 is well absorbed even at very high doses[39].

3. Mechanisms and relationships of vitamin B_6 in various diseases

Vitamin B_6 plays a multifaceted role in various diseases by engaging in multiple biological mechanisms, which contributes to reducing oxidative stress, facilitating neurotransmitter synthesis, modulating inflammatory responses, and maintaining metabolic homeostasis[5]. These actions underline its importance in managing and potentially mitigating conditions such as NDDs, CVD, DM, and cancer. Fig. 4 offers an overview of these mechanisms, illustrating how vitamin B_6 interacts with different pathways in both human and mouse models. Its involvement extends to inhibiting inflammatory factors, enhancing antioxidant enzyme activity, promoting endothelial repair, and regulating metabolic pathways, among other functions. However, further research is essential to better understand its therapeutic potential, particularly in addressing gaps in knowledge brought to light by new challenges like the COVID-19 pandemic.

3.1 Anti-cancer mechanism of vitamin B_6

The significance of vitamin B_6 lies in its relationship between intake, blood levels, and cancer risk[40]. Further attention should be given to vitamin B_6 as a potential strategy for reducing the risk of gastrointestinal cancers. In particular, dedicated randomized controlled trials (RCTs) are needed, potentially involving participants selected based on identified risk factors and blood levels of PLP. Additionally, blood levels of PLP should be considered a novel cancer risk factor and utilized as a biomarker for cancer susceptibility in large-scale screening programs. Furthermore, PLP could be assessed in conjunction with other biomarkers, such as germline polymorphisms involved in carcinogenesis[41]. Given these findings, the current recommended daily dose of vitamin B_6 (1.5 mg) may need to be adjusted upwards. Consumers should be encouraged to adopt diets rich in this vitamin.

Evidence[42] suggests that low vitamin B_6 is associated with an increased risk of lung cancer as assessed by circulating PLP. Since PLP levels may be influenced by parameters other than vitamin B_6 status, a functional vitamin B_6 marker which is the ratio of hydroxykynurenine to xanthurenic acid (H3L) was analyzed. It was further verified that high levels of H3L indicate impaired vitamin B_6 function, which is associated with an increased risk of lung cancer.

Cell-based studies[43] have shown that high levels of vitamin B_6 can suppress the growth of certain cancer cells, which suggest that supraphysiological doses of vitamin B_6 may inhibit tumor growth and metastasis. However, early animal study[44] indicated that moderate doses of dietary vitamin B_6 could suppress azoxymethane-induced colon tumorigenesis in mice. Epidemiological studies[45-47] also support an inverse relationship between vitamin B_6 intake and the risk of colon cancer. Potential mechanisms underlying the preventive effects of dietary vitamin B_6 include the suppression of cell proliferation, oxidative stress, nitric oxide (NO) synthesis, and angiogenesis.

Evidence from *in vitro* and animal studies[48-49] indicates that impaired methyl group metabolism can influence cellular differentiation in the pancreas and contribute to toxic damage in ways that enhance the pathogenesis of pancreatic diseases and carcinogenesis. In terms of human studies of male smokers[50-53], serum levels of PLP (the coenzyme form of vitamin B_6) was inversely associated with risk of pancreatic cancer.

Vitamin B_6 exhibits significant potential in cancer prevention and risk reduction through its diverse mechanisms, including suppression of oxidative stress, regulation of methylation pathways, modulation

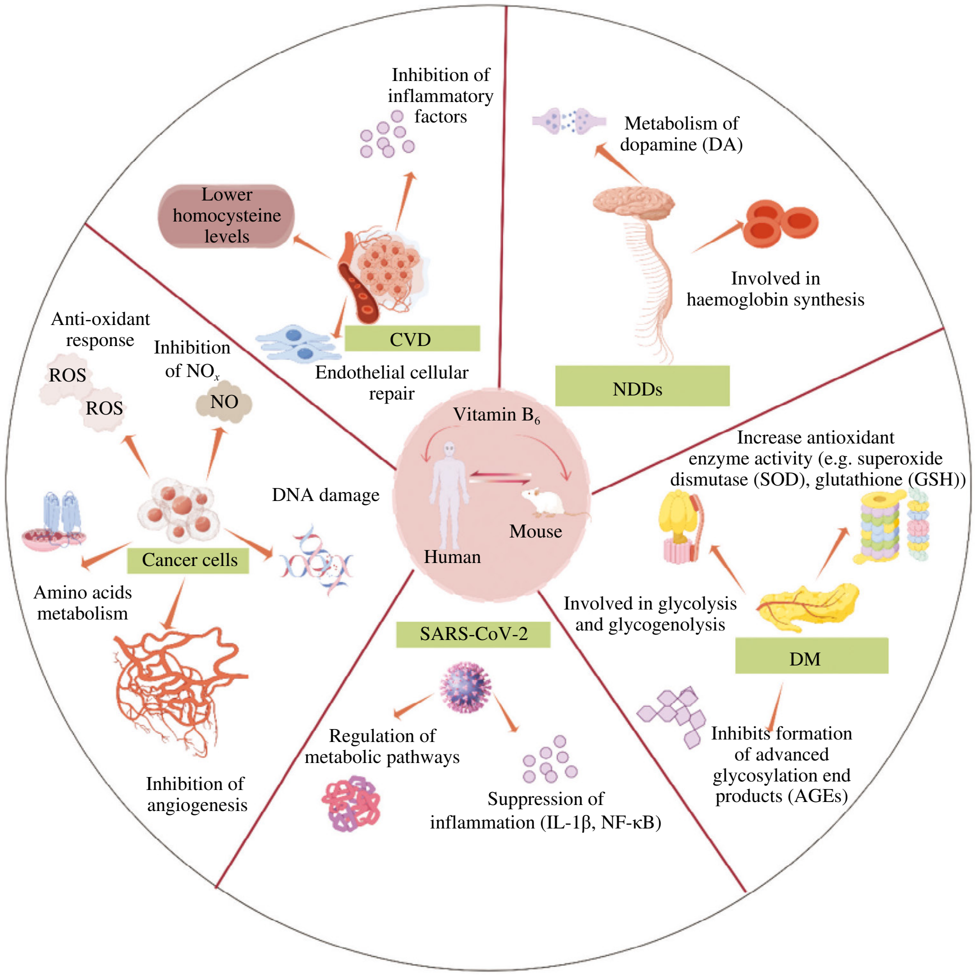

Fig. 4 Overview of the different mechanisms of action of vitamin B_6 in disease (human and mouse). ROS, reactive oxygen species.

of inflammatory responses, and inhibition of cancer cell proliferation and angiogenesis (Table 1). Its inverse association with cancer risk, observed across various types such as gastrointestinal, lung, pancreatic, and breast cancers, underscores its role as a protective factor. Notably, PLP, the active form of vitamin B_6, stands out as a promising biomarker for cancer screening and risk stratification, particularly when combined with other biomarkers such as genetic polymorphisms. Nevertheless, substantial heterogeneity across existing studies and the limited availability of RCTs highlight the need for further research to confirm these findings. Future investigations should prioritize establishing optimal dosing strategies, integrating vitamin B_6 with complementary nutrients for synergistic protective effects, and evaluating its preventive potential in high-risk populations. These efforts may lead to more targeted dietary guidelines and clinical applications for leveraging vitamin B_6 in cancer prevention and management.

3.2 *Inflammatory mechanism of vitamin B_6*

The maintenance and replenishment of the immune cell population rely on a sufficient supply of energy and nutrients. A well-balanced diet is crucial for the proper functioning of the immune system, especially for the active and rapidly proliferating immune cells[54-55]. The detrimental effects of inadequate nutrition on the body's ability to combat infections are well-established.

Several recent studies[56-60] have highlighted the potent anti-inflammatory properties of vitamin B_6. For instance, Selhub et al.[58] investigated the effects of mild vitamin B_6 deficiency and moderate supplementation on colonic inflammation severity in mice by administering varying doses of PN hydrochloride. The results showed that both low and high plasma levels of PLP were associated with the suppression of molecular markers (tumor necrosis factor α (TNF-α), IL-6, interferon-γ, cyclooxygenase-2, and inducible nitric oxide synthase) and histologic markers of colonic inflammation. These findings suggest that vitamin B_6 supplementation could be explored as a therapeutic approach for inflammatory bowel disease. Additionally, Shan et al.[61] supported the idea that vitamin B_6 supplementation can prevent lung inflammation by inhibiting macrophage activation through the 5′-AMP-activated protein kinase-docking protein 3 (AMPK-DOK3) pathway. Vitamin B_6 administration[56] was found to enhance AMPK phosphorylation, thereby increasing lipopolysaccharides (LPS)-induced inflammation while simultaneously reducing the production of TNF-α and IL-1β *in vitro*. In a study by Sakakeeny et al.[62], the PLP levels of 2 229 patients were evaluated, revealing that individuals with chronic inflammation had the lowest vitamin B_6 levels. Furthermore, research has shown that individuals with type 2 diabetes and CVD tend to have lower vitamin B_6 levels. Insufficient vitamin B_6[63] can adversely affect the immune system by disrupting the Th1-Th2 balance and promoting an exaggerated Th2 response, potentially leading to allergic reactions.

Table 1
Summary of vitamin B_6 interventions and their effects in animal and human studies.

Cancer type	Subjects	Diet/supplement	Duration	Performance effects	Potential mechanism	Reference
Lung cancer	Animal (mouse)	Vitamin B_6 benzoyl hydrazone platinum (II) complex (1.6 μmol/kg)	30 days	The inhibitory effect of vitamin B_6-Pt1 on A549 xenograft tumors was 81.5%	Enhanced cellular uptake via vitamin B_6 interaction with cell membrane, and inhibition of efflux pumps (e.g., P-glycoprotein)	[44]
Lung cancer	Human (cohort study)	Lower plasma vitamin B_6 status	/	Increased risk of lung cancer due to impaired vitamin B_6 status	1) Impaired vitamin B_6 metabolism affecting DNA repair mechanisms; 2) reduced ability to mitigate oxidative stress; 3) altered immune function linked to low vitamin B_6 levels	[45]
Colorectal cancer	Human (case-control study)	High dietary intake of B vitamins (including B_6)	One year	Lower risk of colorectal cancer with higher intakes of B vitamins and methionine	1) B vitamins involved in DNA methylation and repair; 2) methionine metabolism impacting tumor suppression; 3) role of B vitamins in reducing oxidative stress	[46]
Non-small cell lung cancer	Human (retrospective study)	Serum levels of vitamin B_6	Retrospective analysis	Positive correlation between higher serum vitamin B_6 levels and intrapulmonary lymph node or pleural metastases	1) Vitamin B_6 involved in immune system regulation; 2) potential role in tumor progression and metastasis through immune modulation; 3) influence on inflammatory pathways and cell signaling	[47]
Pancreatic ductal adenocarcinoma	Animal (mice/cells)	Vitamin B_6	5 weeks	Inhibition of NK cell-mediated antitumor immunity in the tumor microenvironment	1) Vitamin B_6 competes with other factors in the tumor microenvironment, reducing NK cell function; 2) altered immune response against tumor cells	[48]
Acute myeloid leukemia	Animal (mice)	Vitamin B_6	12 days	Evidence of vitamin B_6 addiction, supporting AML cell survival or proliferation	1) Vitamin B_6 involvement in key metabolic pathways (e.g., amino acid metabolism); 2) increased reliance on vitamin B_6 for DNA synthesis and cell division in leukemia cells	[49]
Colorectal cancer (stage I–III)	Human (stage I–III)	Higher vitamin B_6 status (dietary intake or supplementation)	3.2 years	Improved survival rates in patients with stage I–III colorectal cancer associated with higher vitamin B_6 levels	1) Vitamin B_6's role in immune system modulation; 2) antioxidant properties reducing inflammation and oxidative stress; 3) supporting DNA repair and cellular metabolism	[50]
Kidney cancer	Human (case-cohort study)	Dietary intake	12 months	Higher circulating concentrations of vitamin B_6 were associated with improved survival and prognosis in kidney cancer patients	1) Vitamin B_6 may play a role in modulating immune response and reducing oxidative stress, which can influence cancer progression; 2) adequate vitamin B_6 levels might contribute to better tumor control, enhancing overall survival rates	[51]
Primary liver cancer	Human	The sodium cantharidinate/vitamin B_6 (50 mL, once daily)	2 weeks	The combination of sodium cantharidinate and vitamin B_6 showed therapeutic effects, improving liver function and reducing tumor size in primary liver cancer	Vitamin B_6 may enhance the immune response, reduce oxidative stress, and improve liver function in cancer patients	[52]
Gastric cancer	Human (epidemiological studies)	Vitamin B_6 intake (dietary or supplementation)	/	Higher intake of vitamin B_6 was associated with a reduced risk of gastric cancer, with stronger effects observed in some populations	1) Vitamin B_6 may have anti-inflammatory effects that reduce gastric cancer risk; 2) it could regulate genes involved in cell cycle control and apoptosis	[53]

Note: / Indicates not stated in the literature.

Overall, numerous studies[62-64] have illustrated the importance of vitamin B_6 in various aspects of immune function, including infection prevention, immune response, intestinal immunity enhancement, and the augmentation of NK cell cytotoxic activity.

Vitamin B_6 has been shown to exert anti-inflammatory effects through its involvement in the NLR family pyrin domain containing 3 (NLRP3) inflammasome and nuclear factor κ-light-chain-enhancer of activated B cells (NF-κB) pathways, as highlighted in several studies[63-65]. It has been reported[63] that vitamin B_6, particularly in its active form PLP, can suppress inflammation by inhibiting the activation of NF-κB and the NLRP3 inflammasome, which is responsible for caspase-1 activation. This effect is mediated by PLP's ability to block the phosphorylation of transforming growth factor-β activated kinase 1 (TAK1) and c-Jun N-terminal kinase (JNK), which are activated by LPS, and by interfering with the inhibitor of κB kinase-inhibitor of κB α (IKK-IκBα) pathway. Supplementation with vitamin B_6 (PL and PLP) has been shown to significantly reduce the NLRP3 inflammasome's response to various stimuli in LPS-primed macrophages, potentially contributing to the regulation of heart health and reducing the production of pro-inflammatory cytokine IL-1β. Additionally, vitamin B_6 reduces mitochondrial ROS production in peritoneal macrophages, which plays a critical role in NLRP3 inflammasome activation. This suggests a novel mechanism by which vitamin B_6 may also act as an antioxidant. Further, an animal model study[66] demonstrated that vitamin B_6 injection reduced IL-1β production and protected mice from LPS-induced endotoxic shock. The NLRP3 inflammasome is activated by the P2X7 receptor (P2X7R) through enhanced ROS production, and this pathway has implications in myocardial infarction and myocardial ischemia/reperfusion injury. Notably, *in vitro* studies[61,66-68] have shown that PLP can directly inhibit the P2X7R at physiological levels. Therefore, vitamin B_6 may play a role in suppressing heart diseases by inhibiting the inflammasome and reducing the production of IL-1β and IL-18.

Vitamin B_6, in its active form PLP, suppresses inflammation by inhibiting key signaling molecules such as TAK1, JNK, IKK, and IκBα, which are activated by LPS. This inhibition reduces the activation of the NLRP3 inflammasome, which is a critical mediator of inflammation, and decreases the production of pro-inflammatory cytokines like IL-1β and IL-18. Furthermore, vitamin B_6 has been shown to reduce mitochondrial ROS production, a key trigger for inflammasome activation, thereby providing an antioxidant effect. Studies also indicate that vitamin B_6 may protect against conditions like myocardial infarction and endotoxic shock by modulating these pathways. Additionally, vitamin B_6 has potential therapeutic applications in lung inflammation, such as acute pneumonia, through the AMPK-DOK3 signaling pathway, suggesting its broad therapeutic potential in inflammatory diseases.

3.3 Vitamin B_6 mechanism for DM

DM is a chronic condition that represents a major public health challenge globally. The International Diabetes Federation forecasts that by 2045[69], over 700 million people worldwide will be affected by diabetes. DM includes a range of metabolic disorders[70] that result in persistent high blood sugar levels due to issues with insulin secretion, action, or both. Chronic hyperglycemia triggers various harmful responses in the body, such as oxidative stress and the formation of advanced glycosylation end products, which lead to structural and functional changes in blood vessels. These changes contribute to the development of complications in vital organs, including the heart, nerves, eyes, and kidneys[71-72]. Given the impact of lifestyle and dietary changes on diabetes, early intervention is critical to prevent more severe complications.

Vitamin B_6, in its active form as the coenzyme PLP, plays a critical role in various enzymatic reactions, including gluconeogenesis and glycogenolysis[3,73]. There is significant evidence linking vitamin B_6 to diabetes and its associated complications, such as gestational diabetes, diabetic pain, and neuropathy[74-77]. Interestingly, research suggests[78] that vitamin B_6 deficiency can both result from and contribute to the development of diabetes, highlighting its potential bidirectional relationship with the disease, as presented in Fig. 5.

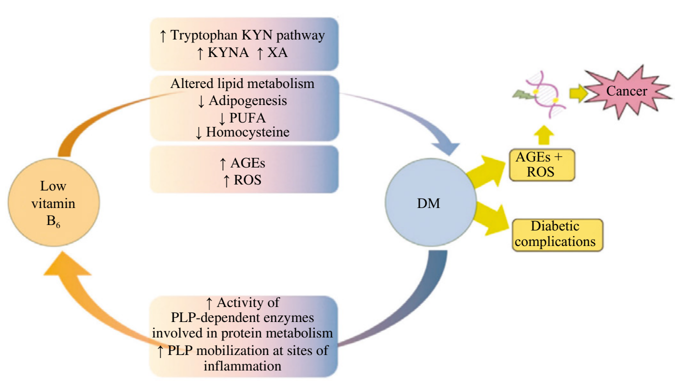

Fig. 5 Schematic representation of the main mechanisms and pathways involved in inferring the association between vitamin B_6 and DM[78]. KYN, kynurenine; KYNA, kynurenic acid; XA, xanthurenic acid; PUFA, polyunsaturated fatty acid.

Several animal studies[75,79-84] have demonstrated a correlation between vitamin B_6 deficiency and the development of diabetes. For instance, research by Abdullah et al.[81] revealed that treatment with PM (at doses of 10 and 15 mg/kg body weight) led to significant recovery in various parameters, including antioxidant enzymes and oxidative stress markers such as SOD, catalase, GSH, lipid peroxidation, and protein oxidation. Additionally, 2′,7′-dichlor odihydrofluorescein diacetate (DCFH-DA) microscopy showed improvements in these markers compared to the diabetic group. Histopathological analysis of liver and kidney tissues, along with assessments of cellular DNA damage and nuclear aberrations, further supports the protective effect of vitamin B_6 against diabetes. These findings align with previous research[83] indicating that PM may reduce the incidence of diabetes.

Previous study[85] involving Japanese patients with type 2 diabetes has shown that individuals with a higher intake of vitamin B_6 (2 mg/day) have approximately a 50% lower risk of developing diabetic retinopathy compared to those with a lower intake (0.9 mg/day). This finding suggests that medical nutritional therapy aimed at ensuring adequate vitamin B_6 intake could help prevent the development of diabetic retinopathy in type 2 diabetes patients. Additionally, several systematic reviews[78,86] on vitamin B_6 and diabetes have provided a detailed summary of the molecular mechanisms linking vitamin B_6 levels to diabetic complications. Validation of these findings[87] was conducted using an animal model of diabetes. Results indicated that vitamin B_6 administration protected retinal photoreceptor cells in mice from photodamage, primarily by inhibiting superoxide production and inflammation in microvascular endothelial cells exposed to late-stage glycosylation end products, and by reducing the production of basement membrane proteins involved in capillary detachment in the retina[88].

In conclusion, vitamin B_6 plays a significant role in managing DM and its complications, with research indicating that adequate intake of vitamin B_6 can help prevent or mitigate the effects of DM (Table 2). Animal studies have shown that vitamin B_6 supplementation improves antioxidant defenses, reduces oxidative stress, and protects against tissue damage, particularly in the liver, kidneys, and retina. Human studies support these findings, with higher vitamin B_6 intake associated with a reduced risk of diabetic retinopathy. Vitamin B_6 is involved in metabolic processes such as gluconeogenesis and glycogenolysis, and its deficiency has been linked to both the onset and progression of diabetes. However, more clinical studies are necessary to translate these findings into effective dietary guidelines, especially considering the variability in dietary habits across regions and the impact of age on diabetes prevalence and severity.

3.4 Association of vitamin B_6 with the risk of CVD

Epidemiological studies[93-95] have demonstrated that the consumption of fruits and vegetables is beneficial to cardiovascular health and is negatively associated with the risk of CVD. Further experimental researches[95-96] have explored bioactive substances, such as vitamins, proteins, and trace elements, that favor cardiovascular protection. Notably, previous studies[97-99] have investigated the potential pathophysiological mechanisms underlying the beneficial effects of vitamin B_6 in preventing CVD, including homocysteine lowering, anti-inflammatory effects, endothelial function improvement, and blood pressure regulation, all of which are well-known risk factors for CVD.

Vitamin B_6 is involved in the metabolism of homocysteine, an amino acid associated with an increased risk of CVD. Early animal studies[100-102] found that elevated homocysteine levels caused atherosclerosis in experimental animals. Subsequent research[103] has suggested that adequate vitamin B_6 may help regulate homocysteine levels. For example, a large meta-analysis of prospective studies[94] in patients with homocysteine metabolism defects and elevated serum

Table 2
Effects of vitamin B_6 in DM.

Type of diabetes	Subjects	Diet/supplement	Duration	Performance effects	Potential mechanism	Reference
Gestational diabetes	Animal (mouse)	Low vitamin B_6 (0.5 mg/kg pyridoxine HCl) diet	3 weeks	Increased blood glucose levels and impaired glucose tolerance in pregnant mice; susceptibility varied by strain	Strain-dependent differences in vitamin B_6 metabolism, altered glucose homeostasis, and increased insulin resistance	[89]
Gestational diabetes	Animal (mouse)	Vitamin B_6-deficient (0.5 mg/kg pyridoxine HCl) diet	3 weeks	Impaired glucose tolerance and insulin secretion; onset of gestational diabetes	Disrupted serotonin signaling in pancreatic β-cells due to vitamin B_6 deficiency, impairing insulin production	[75]
Type 2 diabetes	Human	Baseline plasma PLP (vitamin B_6 status)	5 years	Low plasma PLP levels correlated with higher all-cause and cardiovascular mortality	Impaired vitamin B_6 metabolism leads to increased inflammation and oxidative stress, contributing to mortality risk	[47]
Type 2 diabetes	Human (retrospective study)	Dietary vitamin B_6 intake	8-year follow-up	Higher vitamin B_6 intake associated with lower incidence of diabetic retinopathy	Antioxidant properties of vitamin B_6 reduce oxidative stress and inflammation in retinal tissues	[85]
Type 1 diabetes	Human (children)	Vitamin B_6 (200 mg/day)	8 weeks	Significant improvement in endothelial function, measured by flow-mediated dilation	Folate and vitamin B_6 reduce oxidative stress and homocysteine levels, improving vascular health	[48]
Type 2 diabetes	Animal (drosophila)	1 mmol/L PLP	2 days	Reduced DNA damage and improved survival under diabetic conditions	Vitamin B_6 acts as an antioxidant, reducing ROS and protecting DNA from oxidative damage	[79]
Type 2 diabetes	Human	Dietary vitamin B_6 intake	Two 24-h dietary recall interviews	Higher vitamin B_6 intake correlated with lower risk of diabetic retinopathy (DR) and improved prognosis in those with DR	Vitamin B_6 reduces inflammation and oxidative stress, key contributors to retinal damage in diabetes	[90]
Type 2 diabetes	Human (participants in a prospective cohort study in Shanghai)	Intake of B vitamins (e.g., B_6, B_{12}, and folate)	12 months	Higher joint intake of B vitamins was associated with a reduced risk of developing type 2 diabetes. The protective effect was partially mediated by lower levels of inflammation markers	Adequate B vitamin intake reduces systemic inflammation by decreasing homocysteine levels and oxidative stress, thus improving insulin sensitivity and glucose metabolism	[91]
High-fat diet	Animal (mouse)	Vitamin B_6 (50 mg/(kg·day))	8 weeks	Mice receiving vitamin B_6 supplementation showed improved endothelial function, reduced insulin resistance, and decreased hepatic lipid accumulation compared to those on a high-fat diet without supplementation	Vitamin B_6 mitigates oxidative stress and inflammation, key contributors to endothelial dysfunction and metabolic disturbances. It also regulates lipid metabolism, preventing excessive lipid accumulation in the liver	[92]

homocysteine levels concluded that for every 3 μmol/L reduction in serum homocysteine, the risk of ischemic heart disease was reduced by an average of 16%. The common pathway through which folic acid, vitamin B_6, and vitamin B_{12} contribute to CVD prevention is the reduction of elevated homocysteine levels, which are hypothesized to lead to endothelial dysfunction, promote autoimmune responses, and contribute to the accumulation of inflammatory mononuclear cells in atherosclerotic plaques[93].

As discussed earlier[95], vitamin B_6 may possess anti-inflammatory properties and has the potential to improve endothelial function, i.e. the function of the vessel wall. Healthy endothelial function is critical for cardiovascular health, and several studies[97-98] have shown that vitamin B_6 supplementation may positively influence inflammatory markers and endothelial function. Additionally, vitamin B_6 may play a role in regulating blood pressure. Adequate vitamin B_6 levels are associated with a reduced risk of hypertension, a major risk factor for CVD.

Although some studies[104-105] have suggested the potential benefits of vitamin B_6 in controlling blood pressure, the results have not been entirely consistent. Vitamin B_6, along with other B vitamins, has antioxidant properties that help neutralize free radicals in the body. Oxidative stress is known to be associated with the development of CVD, and these antioxidant effects may contribute to the cardiovascular benefits of vitamin B_6[99]. Longitudinal studies[106] assessing the relationship between vitamin B_6 supplementation and CVD outcomes have been conducted in clinical trials and observational studies, often focusing on dietary intake prior to disease onset.

In conclusion, further research is needed to explore the long-term relationship between vitamin B_6 and CVD, particularly through prospective cohort studies and by assessing dietary intake prior to disease onset. The current understanding highlights that vitamin B_6 plays a key role in regulating homocysteine levels, a risk factor for CVD, and may have anti-inflammatory effects, improve endothelial function, and help manage blood pressure (Table 3). However, the results of studies on vitamin B_6's effects on blood pressure have been inconsistent. Additionally, its antioxidant properties may further contribute to cardiovascular protection by neutralizing oxidative stress. Despite some promising findings, more longitudinal studies are necessary to establish clearer associations and evaluate the specific impact of vitamin B_6 supplementation and dietary intake across different populations and regions. These region-specific analyses are crucial for understanding the full potential of vitamin B_6 in CVD prevention.

Table 3
Effects of vitamin B_6 on CVD.

Subjects	Diet/supplement	Duration	Performance effects	Potential mechanism	Reference
Human	Dietary intakes of vitamin B_6	Two 24-h dietary	Higher intakes of folate, vitamin B_6, and vitamin B_{12} were associated with a lower risk of CVD markers, such as reduced homocysteine levels and improved lipid profiles	These B vitamins reduce homocysteine, a known risk factor for CVD, through methylation processes. Additionally, their antioxidant properties help reduce oxidative stress and inflammation, which are critical contributors to CVD progression	[107]
Human (115 664 participants from a UK population-based cohort)	Dietary intakes of vitamin B_6	24-h dietary	Higher dietary intakes of folate, vitamin B_6, and vitamin B_{12} were associated with a reduced risk of cardiovascular events, including heart disease and stroke. Participants with higher intakes exhibited better cardiovascular outcomes	Folate, vitamin B_6, and vitamin B_{12} lower homocysteine levels through methylation, reducing endothelial dysfunction and inflammation. Their antioxidant properties help mitigate oxidative stress, which is a key driver of CVD progression	[108]
Human	Baseline plasma PLP (vitamin B_6 status)	5 years	Lower serum vitamin B_6 levels were associated with higher all-cause mortality and cause-specific mortality, particularly from CVD and cancer. Participants with adequate or higher vitamin B_6 levels had better survival outcomes	Vitamin B_6 reduces oxidative stress and inflammation, which are key contributors to chronic diseases such as CVD and cancer. It also supports metabolic processes and immune function, improving overall health and resilience against fatal conditions	[109]
Human	Dietary vitamin B_6 intake	2 years	Higher dietary intake of vitamin B_6 was associated with a reduced risk of CVD, including lower incidence of heart attacks and strokes	Vitamin B_6 reduces homocysteine levels, a significant risk factor for CVD, through its role in one-carbon metabolism. It also exhibits antioxidant and anti-inflammatory properties, protecting endothelial cells and preventing atherosclerosis	[109]
Human	Dietary intakes of vitamins	19 years	Higher intake of certain vitamins, including vitamin B_6, was associated with reduced all-cause and cardiovascular mortality among individuals with prediabetes	Vitamins like B_6 play critical roles in reducing oxidative stress and inflammation, improving endothelial function, and maintaining glucose metabolism	[110]
Human (elderly men)	Dietary intakes of vitamins	5 months	The potential impact of higher intake of certain B vitamins (e.g., B_6, B_9, B_{12}) in reducing cardiovascular risk factors or improving specific biomarkers like homocysteine levels, which are linked to heart disease risk	Elevated homocysteine levels are a known risk factor for CVD, and B vitamins, particularly B_6, B_9, and B_{12}, are crucial for its breakdown	[79]
Human (women)	Daily intake of a combination pill of 50 mg of vitamin B_6	7.3 years	Folic acid, vitamin B_6 and vitamin B_{12} combination tablets significantly reduced homocysteine	By reducing homocysteine, B vitamins may help protect against the development of CVD and related complications, thus potentially lowering cardiovascular events and mortality rates	[111]

3.5 The relationship of vitamin B_6 and NDDs

NDDs represent a diverse group of neurological disorders that significantly impact millions of individuals worldwide[112-114]. These diseases, including Alzheimer's disease, Parkinson's disease (PD), and amyotrophic lateral sclerosis, are characterized by cognitive, psychiatric, behavioral, motor, muscular, and sensory dysfunction due to neuronal loss[114-116]. The incidence of NDDs increases with age, leading to their classification as geriatric diseases[117-118].

PD is primarily characterized by the degeneration of dopamine neurons, leading to decreased dopamine levels in the striatum. Levodopa (*L*-DOPA) is considered the standard treatment for PD and has been in use for over 50 years[116,118]. *L*-DOPA is converted to dopamine by the enzyme DOPA decarboxylase, and PLP, the active form of vitamin B_6, is a crucial cofactor in this reaction[119-122]. However, while levodopa is effective, it can cause adverse effects, such as nausea, vomiting, dizziness, and hypotension[123]. To counteract these effects, levodopa is often combined with carbidopa, which prevents the peripheral conversion of *L*-DOPA to dopamine by binding to PLP[124-126]. Unfortunately, long-term use of carbidopa-levodopa can lead to vitamin B_6 deficiency, particularly in high-dose users, which may exacerbate disease progression.

Several clinical cases[123,127] support the hypothesis that vitamin B_6 deficiency is linked to the progression of PD. For instance, a 78-year-old male patient[127] with PD, treated only with carbidopa-levodopa, developed epileptic symptoms that did not subside until high doses of steroids and intravenous vitamin B_6 were administered. After vitamin B_6 supplementation, the patient's epileptic symptoms resolved. This case suggests that the high doses of carbidopa-levodopa may increase the body's need for vitamin B_6. In another study[128], a 75-year-old male patient with advanced PD, treated with levodopa, developed microcytic hypochromic anemia and low serum vitamin B_6 levels. Supplementation with pyridoxal phosphate hydrate improved the patient's condition without worsening PD symptoms[121-122]. Vitamin B_6 plays a key role in synthesizing hemoglobin, and its deficiency has been linked to anemia in PD patients treated with levodopa/carbidopa.

Overall, the use of high-dose levodopa/carbidopa in PD patients can lead to vitamin B_6 deficiency, which may result in adverse effects such as seizures, anemia, and other complications. Clinical cases suggest that vitamin B_6 supplementation can alleviate these symptoms, highlighting the importance of monitoring and addressing B_6 levels in PD patients on long-term carbidopa-levodopa treatment. However, the precise mechanisms through which vitamin B_6 deficiency develops in these patients remain unclear, and further large-scale studies are needed to fully understand the implications and ensure optimal treatment strategies.

3.6 Vitamin B_6 in prophylaxis and therapy of COVID-19

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) has been spreading globally for three years since its identification in

December 2019[129-131]. The cumulative number of confirmed cases worldwide[132] has surpassed 750 million, with the number of deaths exceeding 7 million. Unfortunately, the NeoCoV epidemic continues to spread globally, and some epidemiologists suggest that long-term coexistence between humans and NeoCoV may occur[133-135].

Significant systemic inflammation[136-138] has been shown to negatively impact muscle protein synthesis and increase nutritional demands, which are difficult to meet due to COVID-19-induced loss of appetite, taste, and smell. As a result, the loss of skeletal muscle mass and function (sarcopenia), combined with insufficient intake caused by weakness, depression, and alterations in the gut microbiome, contributes to a high prevalence of malnutrition[139-141]. The impact of malnutrition on the recovery of other systems affected by post-COVID-19 syndrome is significant and warrants critical attention. Dietitians are thus well-positioned to play a pivotal role in the early detection and ongoing management of post-COVID-19 syndrome, with the potential to improve patient outcomes.

The role of lifestyle in long-term antiviral prevention is crucial. However, there has been limited research[142-144] on the antiviral effects of nutrition and exercise, 2 fundamental components of lifestyle prevention. Specifically, further investigation[145] is needed into how functional foods, nutraceuticals, and physical activity can enhance antiviral immune defenses, either independently or in combination. Functional foods and nutraceuticals offer safe and cost-effective strategies for boosting immune function[129,146].

Given the complex interactions and biological systems involved in vitamin B_6, multiple pathways may work together to regulate various aspects of disease. This section reviews and analyzes previous studies to discuss how vitamin B_6 influences key processes in mediating neococcal pneumonia virus and its complications, emphasizing the complex intersections between various biological mechanisms.

Since many individuals do not obtain sufficient vitamin B_6 from natural sources, a question arises: can vitamin B_6 levels be rapidly elevated with a high-dose supplement following COVID-19 infection? Would this approach be both effective and safe? Additionally, a serum threshold for vitamin B_6 that provides protection against COVID-19 has yet to be established.

According to a scientific assessment by the European Food Safety Authority, the European Register of Authorized Health Claims for Nutritional Functions recognizes specific nutrients, including vitamin A, vitamin B (folic acid, vitamin B_6, B_{12}), and essential minerals and trace elements, due to their contribution to the normal function of the immune system. Some researchers[63,147] propose that vitamin B_6 supplementation may enhance immune function in both humans and animals by influencing metabolic pathways that produce immunomodulatory metabolites, such as kynurenine, indole, and sphingosine 1-phosphate. Sphingosine 1-phosphate metabolism regulates lymphocyte transport, endothelial barrier permeability, and cytokine synthesis[148-150]. Furthermore, vitamin B_6 inhibits IL-1β synthesis by blocking NLRP3 inflammasome activation. Recent studies suggest[61,63,148,151-152] that vitamin B_6 may reduce excessive inflammation and cytokine storm syndrome in COVID-19, and this hypothesis has been validated in animal models.

Evidence indicates[63] that vitamin B_6 suppresses inflammation by inhibiting NF-κB activation and NLRP3-mediated caspase-1 activation, as PLP inhibits LPS and IKK-IκBα-induced TAK1 and JNK phosphorylation. However, this study suggests that vitamin B_6 reduces inflammation through AMPK Thr172 phosphorylation and subsequent DOK3 activation. Several studies[61,148,151-152] consistently confirm that AMPK inactivation increases inflammatory responsiveness in mouse macrophages through the activation of NF-κB-dependent gene expression, as well as caspase-1 activation and IL-1β maturation.

In a clinical study from South Korea[143], researchers found that only 6% of COVID-19 patients were deficient in PLP, while 96% had deficiencies in PL or 4-PA. PLP, the active form of vitamin B_6, is a critical cofactor in various inflammatory pathways and is gradually depleted during inflammation. The study[148] also showed that PLP levels did not decrease significantly, possibly because the individuals were less affected, as the study was conducted within seven days of patient admission. Conversely, deficiencies in 4-PA and PL were not measured.

Here, we summarized the available evidences, but the current research on using vitamin B_6 to prevent and treat COVID-19 is limited, indirect, and based on small studies. While there is some evidence supporting the potential benefits of vitamin B_6, clinical studies involving patients who have recovered from COVID-19 and are dealing with related complications are needed to confirm its effectiveness.

4. Conclusions

In summary, addressing nutritional deficiencies, identifying populations with inadequate dietary intake, and implementing feasible, safe, and effective nutrition policies are critical for supporting health recovery after COVID-19. Further research into the underlying nutritional risk factors may help explain why certain populations are more susceptible to COVID-19. Encouraging consistent consumption of recommended vitamin B_6 levels in healthy individuals is also advisable to maintain immune system balance. Given the importance of optimal resistance to SARS-CoV-2, a well-balanced diet supplemented with essential vitamins and micronutrients provides a practical, safe, and accessible therapeutic strategy.

To explore the intersection of vitamin B_6 and disease in greater detail, we conducted a Web of Science search for articles containing the terms "vitamin B_6" and "disease" published over the past 10 years. From these, we selected the 300 most highly cited articles as the key dataset for creating a research hotspot map using VOSviewer software (Fig. 6). The analysis revealed that intersections between these research areas are most commonly associated with age-related groups, including adults, children, and individuals aged 80 and older.

5. Current challenges and outlooks

Despite advancements in research on vitamin B_6 and its role in various diseases, the complex health conditions emerging after outbreaks necessitate a focused approach to investigating causality and underlying mechanisms. While earlier studies have largely explored the modification of specific disease-related metabolites for treating and preventing vitamin B_6 deficiencies, comprehensive mechanistic investigations remain limited. Most research to date has concentrated on the beneficial effects of vitamin B_6 in combination with other dietary components in both animal and clinical models. Greater emphasis should be placed on well-designed clinical and

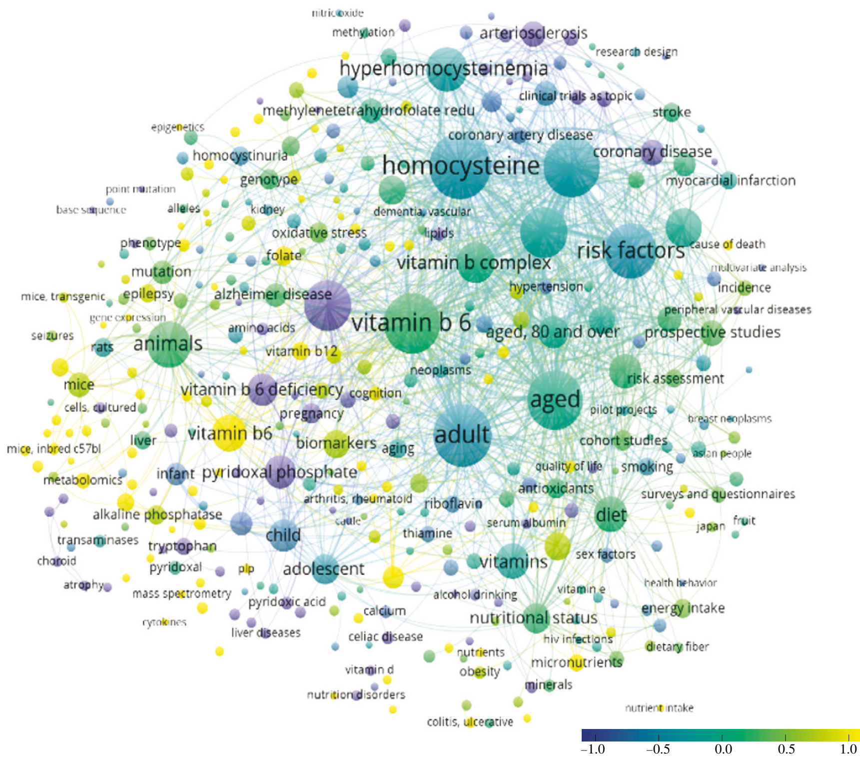

Fig. 6 Research hotspots at the intersection of vitamin B_6 and disease based on VOSviewer software analysis.

translational studies aimed at improving overall well-being and preventing disease progression.

The global spread of COVID-19 has introduced significant inter-individual variations in responses to vitamin B_6 and increased the complexity of regional disease patterns. To address these challenges, future research should focus on: 1) developing country- and region-specific recommendations for vitamin B_6 intake among recovered patients; 2) promoting diversity, equity, and inclusivity through comparative analyses of individuals undergoing rehabilitation and those with other medical conditions; and 3) exploring the use of vitamin B_6 as a standalone supplement or in combination with other nutrients or medications.

Overall, addressing nutritional deficiencies, identifying populations with inadequate dietary intake, and implementing feasible, safe, and effective nutrition policies can support health recovery following COVID-19. Further research and a deeper understanding of underlying nutritional risk factors may help explain the increased susceptibility of certain populations to COVID-19. Encouraging consistent consumption of recommended vitamin B_6 levels in healthy individuals is advisable to maintain immune system balance. Given the importance of optimizing resistance to SARS-CoV-2, adopting a well-balanced diet supplemented with essential vitamins and micronutrients is a practical, safe, and accessible therapeutic strategy.

Achieving these goals requires effective communication and collaboration among international and regional teams to establish a robust evidence base for dietary counseling on the role of immune-supporting nutrients. This coordinated effort will also aid in developing clinical guidelines and effective public health strategies to enhance nutrition and overall well-being.

Conflict of interest

The authors report no conflicts of interest.

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Authors contribution

Yue Zhao: Data organization, formal analysis, initial draft writing, review and editing. Xiaoli Liu: Review, investigative research, funding support. Han Xiao: Review, visualization, project management. Rubing Li: Conceptual design, project management, supervision. Tianyu Zhao: Visualization, initial draft writing, review and editing.

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